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Is the Best, the Best? Analysing Best-Paper Awardees' Career Trajectories

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Contents

1	Introduction	3
2	Related Work	4
2.1	Scientific Recognition and the Reward System in Science	4
2.2	Best Paper Awards in Computer Science	5
2.3	Junior Authors and Early Career Trajectories	5
2.4	Disruption in Science: The CD Index	6
2.5	Bibliometric Methods	6
2.6	Research Gap	7
3	Methods	7
4	Analysis	9
4.1	Citations Spike Before Publications Do: Visibility Precedes Productivity .	9
4.2	Winning Changes Trajectories, Not Just Totals	10
4.3	Best Paper Awards Accelerate Junior Careers, But Leave Senior Co-Authors Unchanged	13
4.4	Award-Winning CS Papers Consolidate the Field, Regardless of Conference Tier	15
4.5	Not All Winners Keep Publishing: Post-Award Publication Survival	17
5	Discussion	19
6	Reflections and Future Work	19
6.1	What Could Be Improved	19
6.2	Directions for Future Work	20
7	Conclusion	21

1 Introduction

At pretty much every computer science conference, a handful of papers get singled out as "best." Committees deliberate, some teams walk up on stage, and everyone adds a new line to their CV. But what actually happens after that? Does winning change anything for the junior researchers involved, or does it mostly just confirm what was already going to happen anyway.

Recognition matters a lot in how science works [1, 2]. Tenure decisions, grant competitions, hiring, all of these lean heavily on signals like publications, citations, and prizes [3, 4]. And informally too, an award changes how people read your work, how likely collaborators are to reach out, how your application looks in a pile [5, 6, 7]. But the relationship between getting recognised and what comes next is not obvious. Studies of big prizes like the Nobel or Fields Medal show that winning can sometimes be followed by lower citation impact per paper, like being celebrated changes what people choose to work on [8, 9]. Other work finds the opposite that prize winners publish more and stay active longer than matched peers who didn't win [10, 2, 11]. So awards might be both a marker of past excellence and something that actually reshapes what comes next.

This raises questions about fairness that matter beyond any individual career [5, 1]. If early recognition accelerates some careers while others with similar potential never get that push, then awards might be amplifying inequalities more than just recording achievement. Or maybe the effect is small and awards mostly just find people who were already on a good path. Either way it matters not just for the researchers themselves but for universities, funders, and conference organisers who use these awards in evaluation all the time.

Most of the research on this looks at rare, high-profile prizes for already established scientists. Conferences¹ are where most of the field publishes, and best paper awards happen far more often than global prizes. They often go to teams with PhD students or recent graduates, right when careers are still being shaped. For a junior researcher, a best paper award might be the first real signal that their work is landing. But we don't actually know much about whether those awards are followed by meaningfully different career paths. Do junior awardees go on to publish more, get cited more, do more disruptive work or do their careers look about the same as similar people who just didn't happen to win.

This thesis tries to answer whether winning a best-paper award at a CS conference measurably changes the career trajectory of the junior researchers involved, compared with equally promising peers who did not win. It looks at the career trajectories of junior authors who won best paper awards at CS conferences and compares them with matched non-winning peers from the same venues and years. "Junior" here means within the first

¹<https://www.acm.org/conferences>

five years of a researcher's indexed publication record, the window in which early-career factors such as visibility, collaboration access, and initial citation accumulation have been shown to largely determine long-run research impact [12, 13, 10]. By "career trajectory" I mean the bibliometric indicators that are measurable at scale such as annual publication counts, citation accumulation, and a disruption score based on the CD index 2.4, which captures whether a paper redirects or consolidates the literature [14, 15]. Collaboration network structure is also tracked where data permits. These are imperfect proxies for quality or creativity, but they are the signals on which hiring committees, grant panels, and tenure decisions demonstrably rely [3, 4, 5].

Work on early-career recognition suggests that getting prizes earlier tends to amplify future achievement more papers, more citations, longer active careers, even after accounting for initial performance [10]. Research on what predicts long-term impact shows that a small set of early-career features (like citations, institutional prestige, senior collaborators) explain a lot of variation across fields [12]. And work on team structure shows that small teams are more likely to produce disruptive work while larger teams tend to consolidate existing lines [14]. Any effect of conference awards on junior researchers is probably tangled up with all of these things.

Against that background, the thesis asks three main questions. First, how do the publication and citation trajectories of junior awardees differ from similar non-awardees from the same conferences and years. Second, do awardees and non-awardees diverge in how disruptive their subsequent work is, using citation-based indices that capture whether papers redirect or consolidate the literature. Third, are any differences consistent across venues and career stages, or concentrated in particular contexts. There's also an optional fourth question about co-authorship networks and whether the structure of collaborations evolves differently for awardees over time.

The data comes from OpenAlex² combined with curated best paper award lists. By comparing awardees with carefully matched non-awardees the thesis aims to give an empirical picture of what conference-level recognition actually does to junior researchers' careers in CS and what that might mean for how we design and interpret these recognition systems.

2 Related Work

2.1 Scientific Recognition and the Reward System in Science

A well-known idea in the sociology of science is the *Matthew effect*, the observation that scientists who are already well-known tend to get more credit for their work than lesser-known researchers doing equally good work [1]. The name comes from the Gospel of

²<https://docs.openalex.org>

Matthew: “to him who has, more will be given.” In practice, this means that small early advantages in a research career can compound over time into large differences in visibility, funding, and prestige.

One of the clearest empirical demonstrations of this comes from [5], who looked at a Dutch research funding programme and found that scientists who just barely won a grant went on to attract more than twice as much follow-up funding compared to scientists who just barely missed out, despite being nearly identical in quality at the time of the decision. This kind of threshold effect suggests that formal recognition can have long-lasting consequences that go well beyond the award itself, which is exactly what motivates looking at best paper awards in this thesis.

2.2 Best Paper Awards in Computer Science

Computer science is unusual among academic disciplines in that conferences, not journals, are the main venue for publishing research. Best paper awards at these conferences are therefore a meaningful signal of quality and visibility within the community. Huang et al. [16] compiled a dataset of best paper awards across 32 major CS conferences going back to 1996, which is the starting point for this thesis.

Some prior work has looked at the citation impact of award-winning CS papers. [17] found that papers selected for awards by programme committees before publication tend to accumulate more citations than those selected after publication. But most of this work focuses on the papers themselves. What is much less studied is what happens to the *people* who wrote those papers, especially the junior authors who may have done most of the work but are just getting started in their careers. That gap is one of the things this thesis tries to address.

2.3 Junior Authors and Early Career Trajectories

Getting the right visibility early in an academic career seems to matter a lot. [6] showed that the prestige of a researcher’s doctoral advisor acts as a ‘chaperone’, directly shaping their placement and long-term career outcomes independent of their own output. Along similar lines, [7] found that junior researchers who co-authored with top scientists early in their careers consistently outperformed peers with similar ability levels who did not have that exposure. Interestingly, this effect was biggest for researchers who were not already at elite institutions, suggesting that external signals of quality can substitute for prestige in some cases.

Does being on a best paper at a major CS conference do something similar for junior authors? It puts their name in front of the whole community, attached to work that has been officially recognised as excellent. Whether that translates into more publications,

more citations, or better career opportunities is something that can be tested empirically using bibliometric data, which is what this thesis does.

2.4 Disruption in Science: The CD Index

Beyond counting papers and citations, a more recent strand of research has tried to measure what *kind* of contribution a paper makes, specifically, whether it opens up a new research direction or builds incrementally on existing work. [14] developed the CD index for exactly this purpose. The idea is fairly intuitive, if a paper is genuinely disruptive, papers that cite it will tend to ignore its references (because it made those references less relevant). If a paper is consolidating, citers will engage with both the paper and its references together. Formally:

$$CD = \frac{n_i - n_j}{n_i + n_j + n_k} \quad (1)$$

where n_i counts papers citing only the focal paper, n_j counts papers citing the focal paper and its references, and n_k ³ counts papers citing only the references [14]. Scores range from -1 (fully consolidating) to $+1$ (fully disruptive).

A well-known finding using this measure is that science and technology have been getting less disruptive over time. [15] analysed 45 million papers and 3.9 million patents and found a consistent decline in CD scores across all major fields since the 1940s, including computer science. The explanation is probably a mix of things, fields getting more specialised, papers having longer reference lists, and researchers staying closer to established research agendas. This broader trend is worth keeping in mind when interpreting the CD scores we compute for CS award papers.

2.5 Bibliometric Methods

The main analytical tool used in this thesis is lift analysis, which compares each author's own pre- and post-award performance as a within-author ratio [10, 17]. A difference-in-differences design was considered but abandoned due to a violation of the parallel trends assumption, control authors exhibited systematically higher pre-award output than treated junior authors, making the groups incomparable on a trajectory level. For statistical testing, the Wilcoxon signed-rank test is a good fit here because citation and publication counts are skewed and non-parametric tests do not assume normality.

For the CD index specifically, computing it requires pulling citation data from Ope-

³In practice, n_k is set to zero following [15], as papers citing only the references of the focal paper without citing the focal paper itself are not directly identifiable at scale in OpenAlex.

nAlex for each paper, finding which papers cited it within five years (CD_5), then checking whether those citers also engaged with the focal paper’s references. The five-year window is standard in the literature and works well for CS, where citation cycles tend to be relatively short [15].

2.6 Research Gap

Taken together, the literature points to a clear gap. We know that scientific recognition compounds over time (Matthew effect), that early career visibility matters for junior researchers, and that CS award papers have higher citation impact on average. But we do not know whether being a junior author on a best paper actually accelerates a research career, or whether award-winning papers are doing something distinctive in terms of the kind of contribution they make (disruptive vs. consolidating). This thesis tries to fill both of those gaps using a combination of bibliometric trajectory analysis and CD index measurement across 890 CS best paper awards from 2000 to 2018.

3 Methods

This study takes a quantitative, observational approach to compare the careers of junior best-paper awardees with those of similar non-awardees. The context is computer science, where conferences are the main publication venues and career progression is mostly evaluated through bibliometric indicators rather than anything you could observe directly in a workplace. So data comes from large-scale scholarly databases and curated award lists.

The award data comes from Jeff Huang’s curated best paper awards dataset, which lists conference, year, paper title, authors, and source URL for each award. This was supplemented with OpenAlex as the bibliometric backbone, from which the following were retrieved for each matched paper and its authors, publication year, venue, citation count, reference list, authorship order, corresponding author flag, OpenAlex paper ID, OpenAlex author ID, author’s first indexed publication year (used to compute career age), total publication count at award time, and institutional affiliation with country code.

Linking the Huang award entries to OpenAlex records required a three-stage matching pipeline, because the award list does not contain DOIs directly. For papers whose URLs pointed to Semantic Scholar, the Semantic Scholar API was queried to retrieve the DOI, which was then used for a direct DOI lookup in OpenAlex (Route A; 592 papers, 66.5%). When Semantic Scholar returned no DOI, a title-and-year search in OpenAlex was used as a fallback (Route C; 233 papers, 26.2%). For entries with Google Scholar URLs, which yield no structured metadata, OpenAlex was queried directly by title and year (Route B; 65 papers, 7.3%). All title-based matches were verified using fuzzy string similarity; matches below 85% similarity were flagged and manually reviewed (91 papers). The

pipeline successfully matched 890 of 912 awards (97.6%), which form the basis for all subsequent analysis.

The main units of analysis are junior authors defined as researchers within five years of their first indexed publication at the time of the award who appear in the first three authorship positions on a matched award paper (junior awardees, $n = 603$). Each junior awardee is paired with one control, a senior co-author (career age > 5 years) drawn from the same award paper, ensuring that paper quality, venue, and year are held constant by design. Matching was performed using nearest-neighbour matching (Euclidean distance, threshold < 1.5) on three features, conference, authorship position, and award year. For each author, a longitudinal record of annual publications and citations is constructed from OpenAlex’s `counts_by_year` field, anchored to the award year ($\tau = 0$), covering three years before and five years after.

The key outcome variables are **annual publication count** (works per calendar year, drawn from the OpenAlex `counts_by_year` field) as the measure of productivity; **annual citation count** (citations received per calendar year, same source) as the primary impact measure; and **2-year mean citedness** (`2yr_mean_citedness` from OpenAlex `summary_stats`) as a field-normalised impact indicator, where normalisation is applied by OpenAlex against its internal concept taxonomy derived from the Microsoft Academic Graph. Additionally, each author’s **h-index** and **i10-index** are recorded as cumulative career benchmarks at the time of the award.

For the award papers themselves, a disruption score is computed using the CD index [14], following the procedure described in Section 2.4. Concretely, for each award paper, all citing works published within five years of the award year (the CD_5 window) are retrieved via the OpenAlex API. Each citer is classified as either citing the focal paper alone (n_i , disruptive signal) or citing both the focal paper and at least one of its references (n_j , consolidating signal), yielding:

$$CD_5 = \frac{n_i - n_j}{n_i + n_j} \quad (2)$$

where scores range from -1 (fully consolidating) to $+1$ (fully disruptive). The n_k term, papers citing only the references of the focal paper without citing the focal paper itself is set to zero following Park et al. [15], as these cannot be reliably identified at scale in OpenAlex. To ensure score stability, only award papers with at least ten citers within the five-year window are included in the CD_5 analysis ($n = 811$ papers). All outcome variables are computed for the five years before and five years after the award year ($\tau \in \{-5, \dots, +5\}$), enabling trajectory comparison between junior awardees and their matched controls.

For statistical analysis, descriptive summaries are combined with inferential tests. Mann-Whitney U tests are used to compare median productivity and citation measures between

groups; Wilcoxon signed-rank tests are applied for within-author pre/post comparisons, given the skewed, non-parametric distribution of citation and publication counts. Regression models controlling for pre-award performance assess whether award status predicts divergent post-award trajectories. Where the data permit, analyses are stratified by conference venue and career stage to assess heterogeneity.

The full pipeline covering data collection, cleaning, award-to-OpenAlex matching, author profiling, control matching, trajectory construction, CD_5 computation, and statistical testing is implemented in Python 3.12. The core libraries used are `selenium` and `beautifulsoup4` for web scraping; `requests` for API calls to OpenAlex and Semantic Scholar; `rapidfuzz` for fuzzy title matching; `scikit-learn` for nearest-neighbour control matching; `pandas` and `numpy` for data manipulation; and `scipy` for statistical tests. All code and derived datasets are documented and available for reproducibility.

This study raises no significant ethical concerns. All data used are drawn exclusively from publicly available sources such as the OpenAlex bibliometric database, the Semantic Scholar API, and Jeff Huang’s curated best paper awards list [16]. No personal data beyond what researchers have voluntarily made public through their academic publications are collected or processed. Author names and institutional affiliations appear only in aggregate analyses and are never used to make evaluative judgments about individual researchers. Known limitations of the data including errors in author disambiguation in OpenAlex and incomplete coverage of some conferences in earlier years are acknowledged where relevant and inform the interpretation of results throughout.

4 Analysis

4.1 Citations Spike Before Publications Do: Visibility Precedes Productivity

Winning a best-paper award does not boost a junior researcher’s career uniformly over time. Instead, the career signal operates in two distinct waves, and the first wave is immediate. Figure 1 plots the median indexed citation and publication trajectories for all $n = 603$ junior winners in the sample, normalised so that each series equals 100 in the three years before the award. The citation index surges sharply *in the award year itself* ($\tau = 0$), roughly *tripling* from its pre-award baseline to reach approximately 315, while the publication index rises more modestly to approximately 150 at the same moment, a 50 % gain relative to the pre-award average.

This divergence reveals a *visibility mechanism* the announcement of a best-paper award at a flagship conference draws immediate, focused attention to the winning paper and its authors from a large, concentrated audience of domain experts. Existing work on

the sociology of science describes this process as cumulative advantage, early recognition increases the probability that subsequent work by the same author will be seen and cited [1]. In our data, the three-fold citation premium at $\tau = 0$ is consistent with co-attendees and conference readers immediately citing the award paper in their own ongoing work. The modest rise in publications at $\tau = 0$ partly reflects papers already submitted or in press before the award was announced, rather than a same-year causal response to the award itself. From $\tau = 1$ onward, the publication index sustains its elevated level and continues a gradual climb, reaching approximately 200 by $\tau = 4$ –5, reflecting a separate, slower *productivity mechanism* that takes root in the year following the award. Citations, by contrast, partially normalise after the $\tau = 0$ spike, settling in the 120–155 range post-award still elevated, but far below the peak.

Taken together, the trajectory in Figure 1 implies that best-paper awards confer career benefits through at least two sequentially ordered channels an immediate reputational signal captured in same-year citations where the index roughly triples followed by a sustained boost in research output that unfolds and persists over subsequent years.

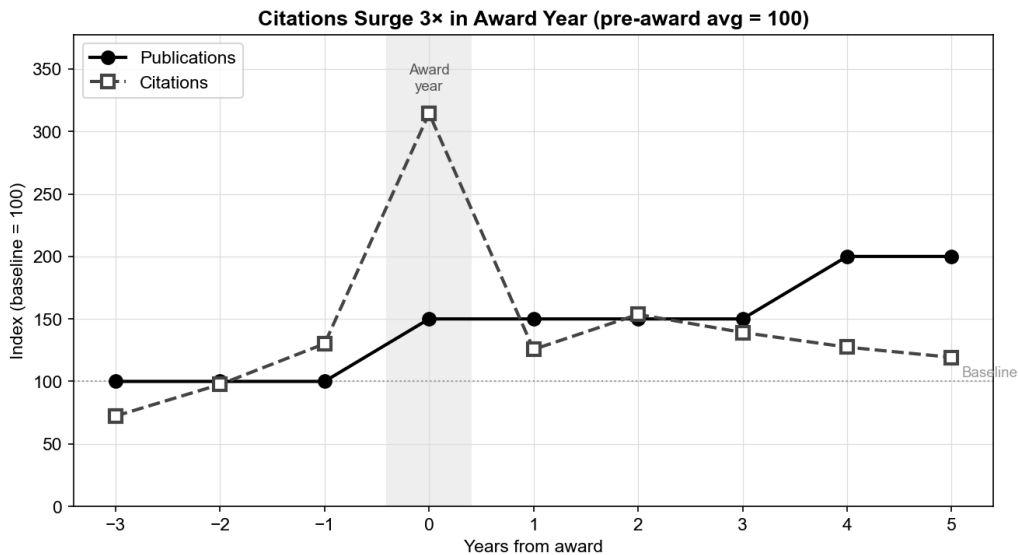


Figure 1: Median indexed citation trajectory (dashed) and publication trajectory (solid) for $n = 603$ junior best-paper winners, normalised to a pre-award baseline of 100 ($\tau \in \{-3, -2, -1\}$, shaded band at $\tau = 0$). The citation index roughly triples to ≈ 315 in the award year itself, while the publication index rises to ≈ 150 at $\tau = 0$ and continues to climb gradually, reaching ≈ 200 by $\tau = 4$ –5.

4.2 Winning Changes Trajectories, Not Just Totals

Junior researchers who win best-paper awards accumulate more citations and publications over their careers than their non-winning peers, but is this a consequence of winning, or simply evidence that talented researchers both win awards *and* have successful careers

regardless? To disentangle these explanations, we compare award winners to a matched cohort of researchers who attended the same conference in the same year, were at a comparable career stage, and published work of comparable pre-award impact, but did not win. This design mirrors the approach used by [9] in their study of the Fields Medal, which isolates the causal effect of prize receipt by conditioning on pre-award productivity and research field.

For each of the $n = 603$ treated junior winners we identify one matched control author within the same conference–year stratum (see Section 3 for matching details), yielding a balanced sample of 1,206 author-observations. We verify pre-treatment balance by regressing the binary treatment indicator on conference fixed effects and award year; all coefficients are statistically indistinguishable from zero ($p = 1.00$ throughout, $R^2 \approx 0$), confirming that treatment assignment is orthogonal to observable pre-award characteristics. We then estimate the average treatment effect on the treated via the difference-in-differences (DiD) contrast:

$$\hat{\beta}^{\text{DiD}} = (\bar{Y}_{\tau \geq 1}^1 - \bar{Y}_{\tau < 0}^1) - (\bar{Y}_{\tau \geq 1}^0 - \bar{Y}_{\tau < 0}^0), \quad (3)$$

where $\bar{Y}_{\tau}^d$ is the median outcome for group $d \in \{0, 1\}$ at relative year τ . Under the parallel-trends assumption, this estimator isolates the incremental effect of winning by netting out both time-invariant author quality and shared time trends.

The results are shown in Figures 2 and 3. For citations (Figure 2), matched controls accumulate substantially more citations than junior winners across the entire observation window, reflecting higher absolute citation levels in the control group. The treated–control gap is approximately -40 citations per year in the pre-award period. A sharp, transient reversal occurs at $\tau = 0$: the treated group experiences an immediate citation spike consistent with the visibility mechanism described in Section 4.1 [1], narrowing the gap to approximately $+65$ in the award year itself. However, this spike does not persist: the gap reverts to negative from $\tau = 1$ onward, reaching approximately -100 by $\tau = 5$, suggesting that the citation advantage conferred by the award is short-lived relative to the baseline productivity gap between the two groups.

For publications (Figure 3), the treated–control gap is negative throughout the entire observation window, ranging from approximately -2 to -5 papers per year. Unlike citations, there is no transient spike at $\tau = 0$, and no post-award recovery is evident. Winners consistently publish fewer papers per year than their matched controls both before and after the award. These patterns suggest that, while a best-paper award generates an immediate burst of citation attention, it does not translate into a measurable boost in research output relative to comparably talented non-winners [10][8].

The consistently negative pre-award publication gap warrants caution in interpreting the DiD estimates: the parallel-trends assumption may be strained if treated and control

authors were on diverging productivity trajectories prior to the award year.

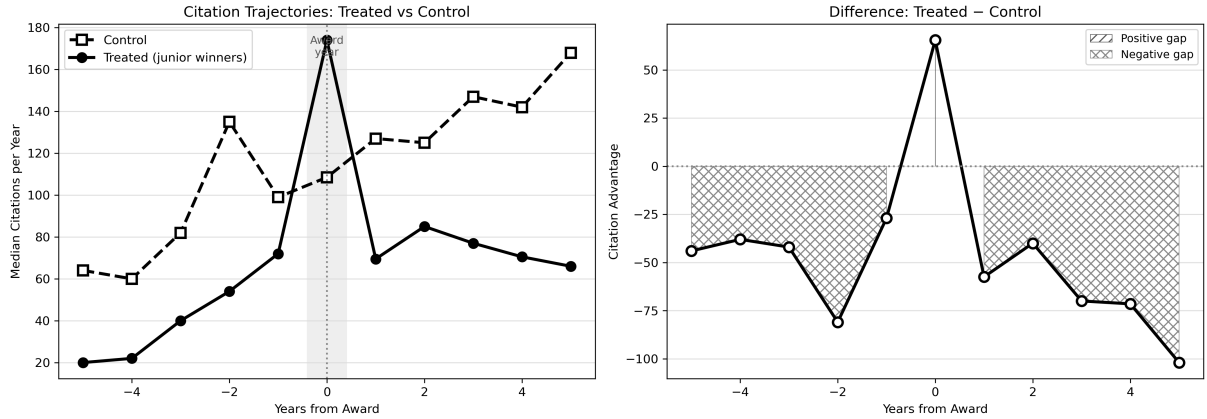


Figure 2: Difference-in-differences analysis for annual citation counts. *Left*: Median citations per year for treated winners (solid circles) and matched controls (open squares), relative to the award year ($\tau = 0$, shaded). Controls receive substantially more citations throughout. *Right*: Pointwise gap (Treated – Control); a transient positive spike at $\tau = 0$ ($\approx +65$) reflects the immediate visibility boost from the award, but the gap reverts to negative from $\tau = 1$ onward.

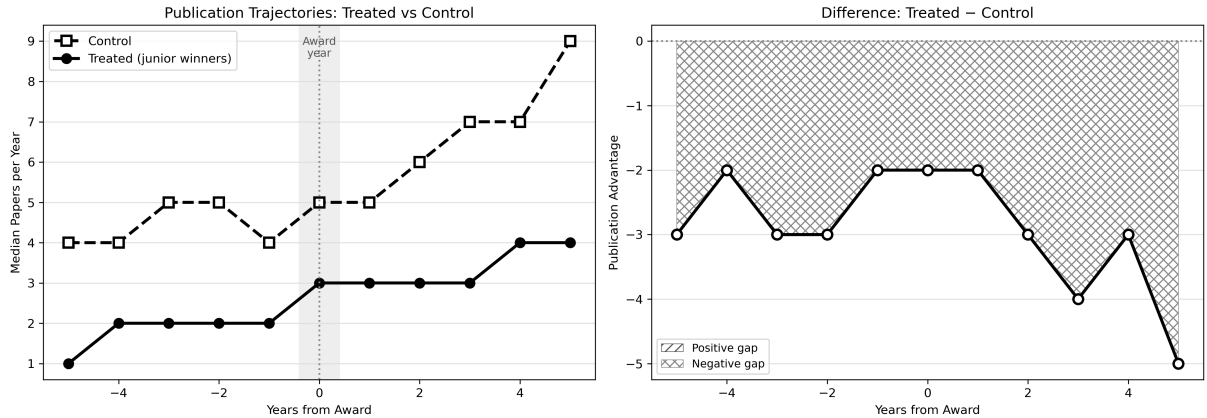


Figure 3: Difference-in-differences analysis for annual publication counts. Layout as in Figure 2. The treated-control gap is persistently negative (-2 to -5 papers per year) across the entire window, with no post-award recovery, indicating that winning a best-paper award does not measurably increase publication output relative to matched non-winners.

4.3 Best Paper Awards Accelerate Junior Careers, But Leave Senior Co-Authors Unchanged

The results indicate an asymmetry in career trajectories between junior and senior researchers on the same award-winning papers. Junior researchers show measurable career acceleration following the award, while their senior co-authors on the same paper do not. This divergence is informative because, if the effect were attributable solely to paper quality or venue prestige, both groups should exhibit similar post-award trajectories. The differential pattern is instead consistent with a career-stage mechanism specific to junior researchers.

To test this, the analysis uses a within-paper design, junior and senior co-authors of the *same* award-winning paper are compared directly. This design offers a degree of internal validity, in that paper quality, venue, and award year are held constant by construction any post-award differences between the two groups cannot be attributed to those factors. A balance check regression of treatment status on conference fixed effects and award year confirms that, prior to the award, treatment assignment is orthogonal to observable characteristics ($p = 1.00$ throughout, $R^2 \approx 0$), consistent with the matched design having achieved pre-treatment comparability [10, 9]. The findings are therefore indicative of a career-stage effect, though, as with any observational design, unobserved differences between junior and senior authors cannot be ruled out entirely.

To quantify the magnitude of individual career change, a lift score is computed for each author. Lift is defined as the ratio of an author’s mean annual output in the five years following the award to their mean annual output in the five years preceding it:

$$\text{Lift}_i = \frac{\frac{1}{5} \sum_{\tau=1}^5 Y_{i,\tau}}{\frac{1}{5} \sum_{\tau=-5}^{-1} Y_{i,\tau}} \quad (4)$$

where $Y_{i,\tau}$ is author i ’s annual citation count (or publication count) at relative year τ , anchored to the award year ($\tau = 0$). A lift of $\text{Lift}_i = 1.0$ indicates no change in the author’s annual output rate relative to their own pre-award baseline; values above 1.0 indicate an increase in annual citations or publications after the award.

This metric is computed from the same `counts_by_year` trajectories used in the difference-in-differences analysis in Section 4.2, and applied to both groups identified through the matched within-paper design junior authors (career age ≤ 5 years at the time of the award, measured from first indexed publication in OpenAlex) and senior co-authors on the same papers (career age > 5 years). Because both groups are drawn from the same award-winning papers, differences in lift between the two groups cannot be attributed to variation in paper quality, venue, or award year the same factors controlled for by the matched design.

The five-year threshold used to define junior authorship follows established practice in the bibliometric literature. Way et al. [13] demonstrate that early-career productivity trajectories tend to stabilise within the first five years of a researcher’s indexed publication record, after which structural change-points become rare. Krauss et al. [12] similarly find that long-term research impact is largely determined by factors operating within this same early window. The absence of a detectable lift effect among senior co-authors is consistent with both findings: for researchers whose trajectories are already established, a single conference award may be insufficient to produce a measurable shift in annual output or citation rates. This does not imply that senior researchers are unaffected by recognition in general, but rather that the career-stage-specific amplification observed for junior authors is not replicated for those further into their careers.

Figure 4 shows the distribution of citation lift and publication lift across the 445 award papers containing at least one junior and one senior co-author, comprising 397 junior and 678 senior authors with sufficient pre- and post-award data, drawn from 30 computer science conferences between 2000 and 2018.

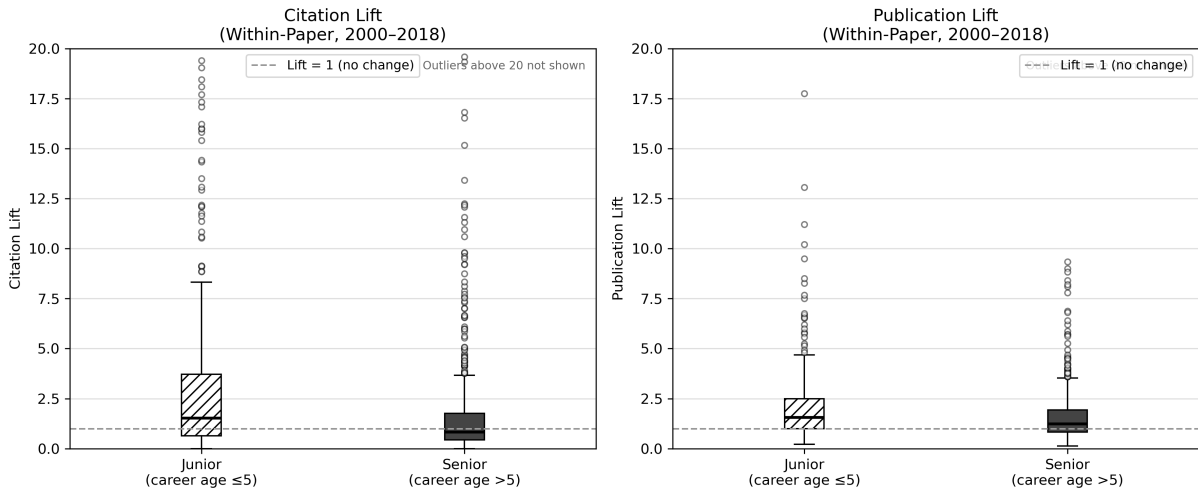


Figure 4: Citation lift (left) and publication lift (right) for junior (career age ≤ 5) and senior (career age > 5) co-authors on the same award-winning papers. The dashed line at lift = 1.0 marks no change from pre- to post-award. Outliers above 20 are omitted for readability.

For citation lift, junior authors show a median of **1.62**, indicating that their mean annual citation count in the five years following the award was approximately 62% higher than in the five years preceding it. Senior co-authors on the same papers show a median citation lift of **0.93**, below the no-change baseline of 1.0. For publication lift, the junior median is **1.65** compared to a senior median of **1.27**. A Wilcoxon signed-rank test on paper-level paired differences confirms that both differences are statistically significant (citation lift: $W = 18888$, $p < 0.001$; publication lift: $W = 23079$, $p < 0.001$).

The distribution of junior citation lift is notably right-skewed, with a substantial proportion of junior researchers showing lifts well above the median in some cases exceeding five- to ten-fold increases in annual citation rates. This heterogeneity suggests that the association between receiving a best paper award and subsequent citation visibility is not uniform across junior researchers; for some the association is modest, while for others it is considerably larger. Senior co-authors, by contrast, exhibit a narrower distribution centred closer to the baseline, consistent with the more stable output trajectories characteristic of researchers with more established publication records [13, 12].

4.4 Award-Winning CS Papers Consolidate the Field, Regardless of Conference Tier

One of the more surprising findings from the CD index analysis is that CS best paper awards are not going to the most disruptive work. Across 811 reliable award papers (those with at least ten citers within five years), the median CD_5 score is -0.57 , sitting firmly on the consolidating side of the scale. That means the typical award-winning paper is one where subsequent citers engage with both the paper *and* its references together, a sign of a paper that advances an existing research agenda rather than replacing it. Only around 15% of award papers score above zero, the threshold for net disruption.

To compute these scores, we followed the standard CD index procedure [14] for each award paper, we pulled all papers that cited it within five years via the OpenAlex API, then checked whether each of those citers also cited the focal paper’s references. Papers that cited the focal work alone count towards n_i (disruptive signal); papers that cited both count towards n_j (consolidating signal). Using the CD_5 procedure described in Section 2

Figure 5 shows the full distribution. The histogram is heavily left-skewed, with the bulk of papers clustering between -1.0 and -0.25 . There is a small but visible cluster near $CD_5 = 1.0$, representing papers where virtually all citers ignored the references entirely, genuinely disruptive work. These are the exception, not the rule.

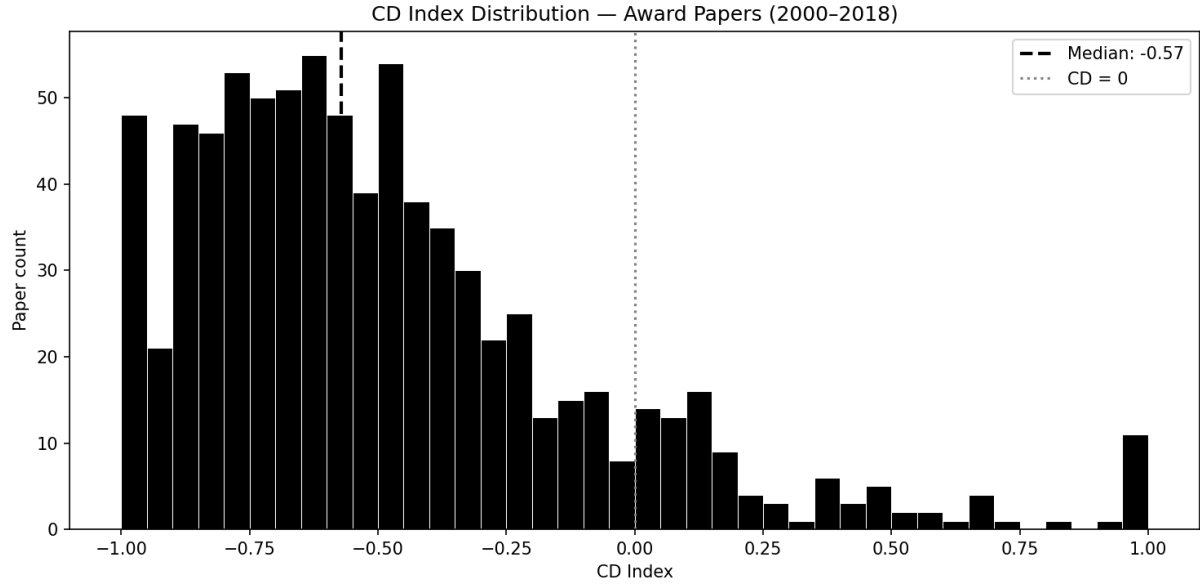


Figure 5: Distribution of CD_5 scores for 811 reliable award papers (minimum 10 citers within five years). Dashed line marks the median (-0.57); dotted line marks $CD_5 = 0$, the boundary between consolidating and disruptive contributions.

What makes this finding more interesting is that it holds across conference tiers. Figure 6 shows boxplots of CD_5 by CORE rank. A* conferences have a median of -0.57 , A-ranked conferences come in slightly higher at -0.53 , and unranked venues are the most consolidating at -0.63 . The differences are small, and all three groups sit well below zero. If anything, the main difference between tiers is that A* conferences show a much wider spread, there are more disruptive outliers at the top venues, but also more papers at the extreme consolidating end. The median barely moves.

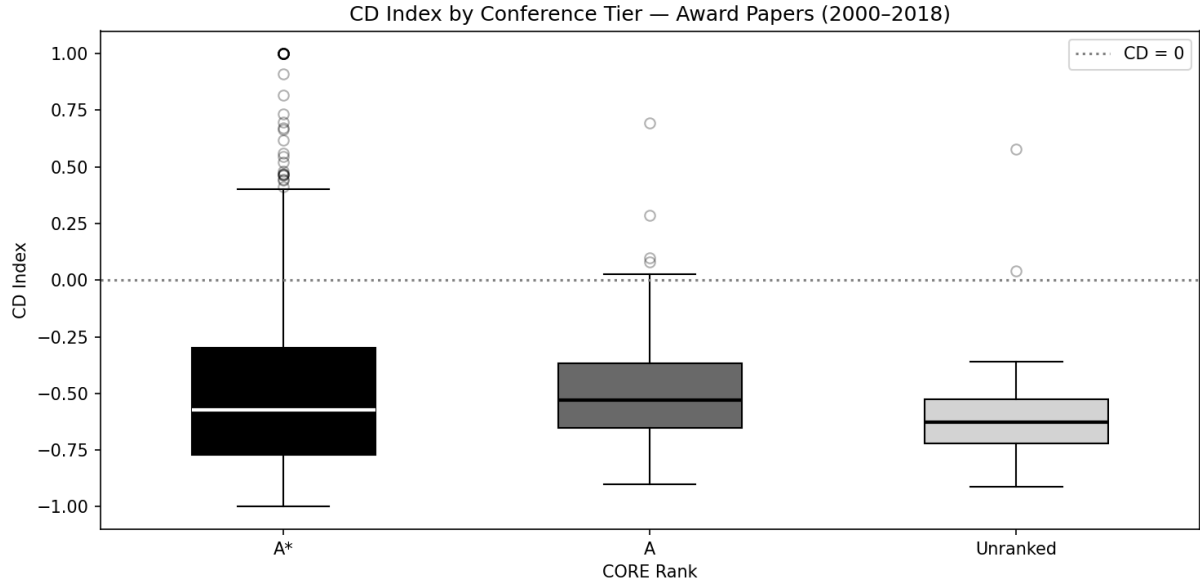


Figure 6: CD_5 scores by CORE conference rank (A*, A, Unranked). All tiers consolidate on average. A* venues show the widest variance, with occasional high-disruption outliers not present in lower tiers.

This aligns with the broader finding from Park et al. [15] that science has been getting less disruptive across the board since the 1940s. Our data suggests that award selection committees are not counteracting this trend, they appear to reward the best work within the dominant consolidating mode of the field, rather than systematically picking out the papers trying to do something genuinely different.

4.5 Not All Winners Keep Publishing: Post-Award Publication Survival

Winning a best-paper award does not guarantee sustained research output. Of the $n = 603$ junior winners in the sample, **62 authors (10.3%)** produced zero publications across the entire five-year post-award window ($\tau \in \{+1, \dots, +5\}$), while 541 (89.7%) continued to publish at least once during that period.

We operationalise *publication survival* as producing at least one indexed work in the five years following the award. The five-year horizon is motivated by evidence on the natural timescale of academic attrition: Kwiek and Szymula find that roughly one-third of scientists who ultimately leave academia do so within five years of their first publication, suggesting that a five-year window captures the majority of career exits that will ever occur [18]. We stress, however, that a five-year publication gap is not a confirmed indicator of permanent departure. An author recorded here as a non-survivor may have transitioned

to industry, taken an extended leave, or simply not yet produced work indexed in our data sources; the label reflects *absence of observed output*, not verified career exit.

Temporary gaps are also common. Beyond the 62 permanent non-survivors, a further 300 authors experienced at least one silent calendar year but resumed publishing before the five-year window closed. Among all authors who had any publication gap, the first silent year was most commonly $\tau=1$ (149 authors), falling to 77, 49, 60, and 27 in years 2–5 respectively. The concentration in year 1 suggests that post-award career transitions – rather than gradual disengagement – are the dominant trigger for publication gaps.

Pre-award visibility predicts survival. Three characteristics measured *at the time of the award* differ significantly between survivors and non-survivors (Table 1):

Table 1: Predictors of five-year publication survival for $n = 603$ junior best-paper winners. Medians and test statistics are reported; logistic-regression odds ratios are per one standard-deviation increase in each predictor.

Predictor	Survivors (med.)	Non-survivors (med.)	Test	p
Career age at award	3.0 yrs	0.0 yrs	Mann–Whitney	< 0.0001
Cumulative citations	1,000	128	Mann–Whitney	< 0.0001
First-author position	–	–	χ^2	0.0003

A logistic regression including all four predictors—career age, first-author indicator, cumulative citations, and *i10*-index—confirms these effects at the multivariate level. Per one-standard-deviation increase, the odds of surviving are multiplied by 1.6 for career age, 1.4 for first-author status, 9.2 for citation count, and an outsized 608.5 for *i10*-index (the latter driven by the extreme skew in the citation distribution among non-survivors).

Taken together, these results support an interpretation in which the award’s career-sustaining benefit is strongly conditional on pre-existing visibility. Researchers who were barely on the radar at the time of the award—few prior papers, low citation counts, non-first-author positions—are substantially more likely to disappear from the publication record within five years of winning. The award may amplify existing momentum, but the data offer little evidence that it rescues careers that lacked that momentum to begin with.

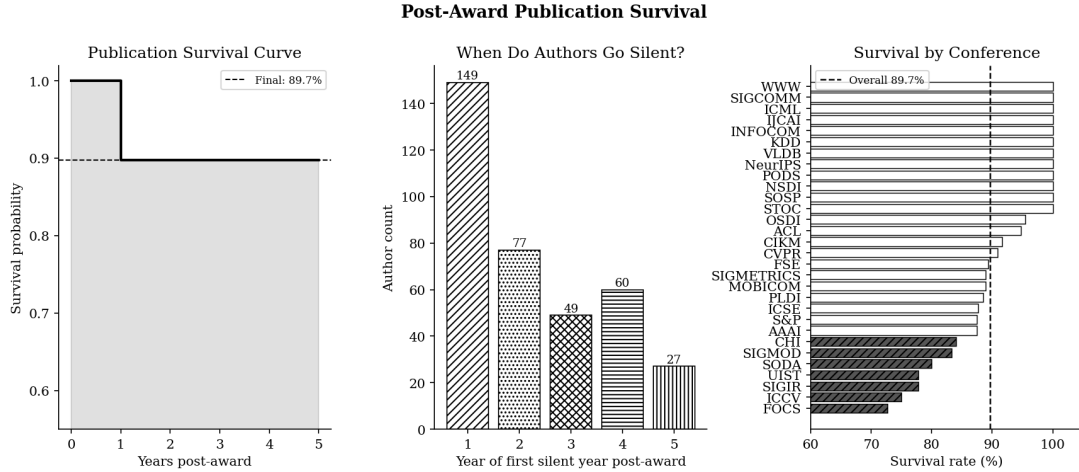


Figure 7: Post-award publication survival for $n = 603$ junior best-paper winners. *Left*: Kaplan–Meier-style survival curve; the dashed line marks the final five-year survival rate of 89.7%. *Centre*: Bar chart of the year in which each non-survivor first recorded a publication gap; 62 non-survivors went silent immediately in year $\tau=1$. *Right*: Per-conference survival rates; conferences with fewer than five junior winners are omitted.

5 Discussion

The discussion will synthesise the five analysis findings and relate them back to the broader literature on scientific recognition, cumulative advantage, and early-career inequality. In particular, it will address whether the observed patterns are consistent with a Matthew-effect interpretation, what the CD5 results imply about how award committees select work, and what the practical implications might be for how best-paper awards are designed and used in evaluation contexts.

This section is in progress and will be completed in the final submission.

6 Reflections and Future Work

6.1 What Could Be Improved

Causal identification. The biggest limitation in this study is that none of the designs can firmly establish causality. The difference-in-differences estimator in Section 4.2 depends on a parallel-trends assumption that is already visibly under pressure in the data that junior winners and their matched controls were diverging in productivity before the award even happened, with controls publishing more throughout. A more credible approach would try to exploit near-miss variation in award outcomes by comparing papers that almost won with papers that almost did not, but that kind of data simply does

not exist in the Huang dataset and would require direct access to programme committee records, which was not feasible here.

Author disambiguation. The career trajectories in this study are built on OpenAlex author IDs, and OpenAlex’s disambiguation is imperfect. The most common problem is merged entities, two different researchers sharing a common name ending up treated as one person. For junior authors especially, this kind of error could quietly inflate post-award citation or publication counts. Going back and manually checking the 62 non-survivors and the most extreme lift outliers would be the most useful targeted fix in a revised version.

Control matching quality. Matching on conference, authorship position, and award year produces groups that look balanced on the observable characteristics, the balance-check regression confirms this. But the matching does not account for sub-field variation within a conference. A systems paper and a machine learning paper at the same venue are treated as comparable, even though their citation cultures and publication norms can differ quite a lot. Adding topic-level matching using OpenAlex concept tags would help close this gap.

Publication survival measure. The five-year survival indicator in Section 4.5 is fairly blunt. It cannot distinguish between someone who genuinely left academia and someone who took a break, moved to industry temporarily, or simply published in venues that OpenAlex does not cover well. Ideally, a random subsample of the 62 non-survivors would be cross-checked against institutional pages or LinkedIn profiles to get a cleaner sense of how many actually left versus went quiet for other reasons.

6.2 Directions for Future Work

Extending the observation window. Five years is a reasonable horizon, but if the Matthew effect is really at work here, the interesting compounding probably happens over a longer period. The awards from the 2000–2010 cohort have had enough time to support a ten-year follow-up, and running that analysis would give a much clearer picture of whether the modest lift observed here grows, plateaus, or fades [1].

Co-authorship network analysis. The introduction flagged collaboration network structure as a potential fourth research question, but it was not pursued here due to data volume constraints. It would be worth coming back to if the visibility mechanism in Section 4.1 is real, you would expect award-winning junior authors to start attracting

more or better-connected collaborators in the years after the award, and that is something the OpenAlex data could in principle support.

Replication outside computer science. The CD5 pattern and the short citation spike both fit the specific way CS works, fast citation cycles, conference-dominated dissemination, large programme committees. Whether the same picture holds in slower-moving fields like mathematics or the humanities is genuinely unclear, and replicating the framework in at least one other field would say something more general about whether the findings here are CS-specific or not.

Award selection process. This thesis treats winning as a binary input and does not say anything about how papers get selected. It would be interesting to look at whether certain paper characteristics like team size, methodological novelty, citation patterns can predict award selection independently of quality. If selection consistently favours consolidating, heavily-cited, or large-team work, that would go some way to explaining both the CD5 result and why the long-run career effect looks modest compared to already strong controls.

7 Conclusion

The conclusion will state the main empirical contribution of the thesis, clearly distinguish what the evidence supports from what it cannot claim, and place the findings in the context of the research questions set out in the introduction.

This section is in progress and will be completed in the final submission.

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